

the idea is that we can trace our path from one point in the domain to another, without picking up our pencil.

Formally, we say a set S in a topological space (X, T) is **connected** if $S = A \cup B$, with both A , and

B being open in X (i. e., $A, B \in T$), with $A \cup B = \emptyset$ **then** either $A = \emptyset$ or $B = \emptyset$.

In other words, in a topological space (X, T) , **a set $S \in T$ is connected if it cannot be written as the disjoint union of two or more non-empty open sets.**

Next, we state a theorem, without proof, as the proof is quite complicated, despite the claim appearing to be quite simple/obvious.

Theorem: In $(\mathbb{R}, d)$, with $\mathbb{R}$, the real-number set, and $d(x, y) = |x - y|$, for any two real numbers x and y , **any interval in $\mathbb{R}$, say $[a, b]$ with $a < b$, is connected.**

Examples and Non-examples:

Example 15.12: In the topology, $(\mathbb{R}, T)$, being induced by mod function $|x - y|$, each of the intervals $]3, 7[,]3, 7], [3, 7[,$ and $[3, 7]$ is connected. In general, any interval—open, closed, semi-open etc.—is connected in this topology.

Example 15.13: For a topology (X, T) , each set in T (by definition, an open set of the topology), is connected.

Example 15.14: None of the following is a connected set in the topology $(\mathbb{R}, d)$, d being induced by mod function $|x - y|$: $]3, 7[\cup]8, 11[$. In general, **union of at least two intervals is not connected**, where intervals are of the form $[a_1, b_1], [a_2, b_2], \dots, [a_i, b_i], \dots, [a_n, b_n]$, in which $a_i < b_i$, and $b_i < a_{i+1}$, for all $i=1, 2, \dots, i, \dots$.

The claim of the example is true for any intervals, not necessarily only closed: even for of the type: $]a, b[, [a, b[,]a, b]$.

15.3.2.3 Compactness of a set in a topological space

In *metric space*, we say a set S is compact set if and only if S is *closed and bounded*. Here, the concept is extended to topological space.

We initiate our discussion with the definition of the concept of **cover**—in a topological space—which is used in the definition of *compactness*.

Cover: In a topological space (X, T) , for a set S , **cover of S is a family F of subsets of X** with the property that $S \subset \bigcup_{B \in F} B$.

Open Cover: A cover is called an open cover if every set B in the family F is an open set, *i. e.* $F \subseteq T$.

Subcover and Finite Subcover of a Cover: For a cover F of the set S , a subcover F' is a family—with $F' \subseteq F$ —which is also a cover of S . If, in addition, F' contains only finitely many sets, we call it a finite subcover.

Compact Set/ Space: A set S in a topological space (X, T) is called compact if every open cover

of S has a finite subcover. If X , itself, is compact, we say X is a **compact space**.

Examples and Non-Examples:

Example 15.15: $\mathbb{N}$, the set of Natural numbers, is not compact in metric space $(\mathbb{R}, d)$, where $d = |x - y|$.

To see this, let us consider open intervals $O_n =]n - 1/4, n + 1/4[$, for $n = 1, 2, \dots$, then

$\mathbb{N} \subset \bigcup_{n=1}^{\infty} O_n$, because, $n \in O_n$ for every $n \in \mathbb{N}$.

But if $\mathbb{N}$ were compact, finitely many of the O_n would cover $\mathbb{N}$. However, each O_n contains only one integer. Therefore, finitely many O_n contain only finitely many natural numbers, and not whole of $\mathbb{N}$. A contradiction.

Check your Progress 3

1) In the metric space $(\mathbb{R}, d)$, where $\mathbb{R}$ is the set of real numbers, and d is the metric induced by mod function $|x - y|$, show that

- i) Each of $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ which is a subset of $\mathbb{R}$, is not connected. And, also, any subset of $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ is NOT connected.
- ii) Any singleton set $\{a\}$ of Euclidean metric space is not connected
- iii) Any finite subset $\{a_1, a_2, \dots, a_n\}$ of Euclidean metric space is not connected.

.....

2) Let $X = \{a, b, c\}$ be a set of three elements. and $T = \{\emptyset, \{a\}, \{a, b\}, \{a, c\}, X\}$. Then show that (X, T) is a topology

.....

3) Let X be any infinite set, and let $T = \{S: S \subseteq X, \text{ and } X - S \text{ is a finite subset of } X\} \cup \{\emptyset\}$, then show that (X, T) is a topology.

.....

4) **Show that** a finite set in $(\mathbb{R}, d)$ —where $\mathbb{R}$ is the set of real numbers, and $d = |x - y|$ —is **compact**.

.....

15.4 LET US SUM UP

In this unit, at the top level, we discussed topics from two very advanced and useful disciplines of Mathematics, viz. Metric Space and Topology. We began with formal definition of each. In the process of defining these, we defined the underlying concepts of open interval/ sphere and closed interval/sphere etc., and discussed their properties.

Within each of these two disciplines, we defined some of their practically useful concepts, including convergence of series, continuity of functions, connectedness and compactness of sets; and discussed their properties and explained some of their applications.

15.5 KEY WORDS

Euclidean Distance d_E between points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in 2-dimensional space R^2 :

$$d_E(P(x_1, y_1), Q(x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Taxicab/ Manhattan distance, d_T between points $P(x_1, y_1)$ and $Q(x_2, y_2)$

$$= d_T(x, y) = |x_1 - x_2| + |y_1 - y_2|.$$

Minkowski distance of order p for R^n , the n -dimensional space over real numbers, is given by

$$\left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p} \quad \text{for order } p = 1, 2, 3, \dots$$

Metric on a Set: For a given non-empty set, say S , and a **function**, (generally, denoted by d): $S \times S \rightarrow R$, the set of real numbers,

which, for x and $y \in S$, satisfies the following conditions, is called a metric

- i) $d(x, y) \geq 0$, (the cost of going from any point to any point is non-negative)
- ii) $d(x, y) = 0$ if and only if $x = y$, i. e.
 - a. $d(x, x) = 0$, for any x in S , (it does not cost to remain at the same point) and
 - b. $d(x, y) > 0$, if $x \neq y$, (the cost of going to a different point is positive),
- iii) $d(x, y) = d(y, x)$ (symmetry property: the cost of going from x to y is the same as the cost of going back to x from y)
- iv) $d(x, z) \leq d(x, y) + d(y, z)$ (transitive property: the cost of going directly from x to z does not exceed the cost of going from x through y to z)

Metric Space	The ordered pair (S, d) , where set $S \neq \emptyset$, and d is a metric on S , is called a metric space.
Open ball/ sphere	For $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n$ a point in the $\mathbb{R}^n$ Euclidean space, open sphere $S(\mathbf{a}, r)$ centred at $\mathbf{a}$ and of radius $r > 0$, a real number, is the following subset of $\mathbb{R}^n$ $S(\mathbf{a}, r) = \{x = (x_1, \dots, x_n) \in \mathbb{R}^n : d_E(x, \mathbf{a}) < r\},$ where d_E is the Euclidean distance on $\mathbb{R}^n$. Closed ball/ sphere in $\mathbb{R}^n$ for n, a natural number is the following subset of $\mathbb{R}^n$ $S^c(\mathbf{a}, r) = \{x = (x_1, \dots, x_n) \in \mathbb{R}^n : d_E(x, \mathbf{a}) \leq r\},$ where d_E is the Euclidean distance on $\mathbb{R}^n$. (The notations S^c is not standard notation for closed sphere)
Open Set	A set S of real numbers is said to be open set, or just open , if for each element a of S , there is a number $\varepsilon > 0$ such that the open interval $]a - \varepsilon, a + \varepsilon[\subseteq S$.
Closed Set	A set $S \subseteq \mathbb{R}$ is said to be a closed if $\mathbb{R} - S$ is open in $\mathbb{R}$.
Open Sphere Closed Sphere of Arbitrary Metric Space	
Open Sphere	For a metric space (X, d) , where set $X \neq \emptyset$, and d is a metric on X , the open sphere with centre a and radius $r > 0$, a real number, is the set
	$S(a, r) = \{x \in X : d(a, x) < r\}.$
Closed Sphere	For a metric space (X, d) , where set $X \neq \emptyset$, and d is a metric on X , the closed sphere with centre a and radius $r > 0$, a real number, is the set
	$S(a, r) = \{x \in X : d(a, x) \leq r\}.$
Open Set of a Metric Space (X, d)	A subset T of X is said to be an open set , if for each element x of T , there is an open sphere $S(x, \varepsilon)$, with $\varepsilon > 0$ such that $S(x, \varepsilon) \subseteq T$.
Closed Set of a Metric Space (X, d)	A set $T \subseteq X$ is said to be a closed if its complement $X - T$ is an open set in X .
Convergence of a Sequence in Metric Space	For (X, d) , a metric space, with $X \neq \emptyset$, and metric d on X , and an infinite sequence $(x_n) = (x_1, x_2, \dots, x_n, \dots)$, of points of X , we say that the sequence (x_n) is convergent if there exists a point x in X such that for each $\varepsilon > 0$, there exists a positive integer k , such that $d(x_n, x) < \varepsilon$; for all $n > k$.
Continuity of a Function in Metric Space	For two metric spaces (X_i, d_i) , for $i = 1, 2$, a function $f: X_1 \rightarrow X_2$ is said to be continuous at point $x = x_0$ of X_1 , if for each $\varepsilon > 0$, there exists a real number $\delta > 0$, such that for all points x in X_1 , satisfying

$$d_1(x, x_0) < \delta \text{ imply } d_2(f(x), f(x_0)) < \varepsilon.$$

Connected Set

A set S in a metric space (X, d) is said to be **connected**, if it cannot be represented as the union of two or more **disjoint** non-empty open sets in the metric space.

Bounded Set and Bounded Metric Space

A subset S of X in a metric space (X, d) is called **bounded** if there exists $r > 0$ such that for all s and t in S , we have $d(s, t) < r$. **Metric space (X, d) is called a bounded metric space** (or d is a bounded metric) if X is bounded as a subset of itself.

Topology

For a given non-empty set X , **a topology on X is the ordered pair (X, T)** , where T is a collection of subsets of X , (i. e. $T \subseteq P(X)$, the power set of X), satisfying the following conditions

- i) the universal set $X \in T$
- ii) the empty set $\emptyset \in T$
- iii) the union, of any number of sets from T , is a set in T ; the intersection, of any *finite* number of sets from T , is a set in T .

*A topology (X, T) is also called a **Topological Space**.*

Open Set in a Topology

Each set in T is called an **open set** of the topological space (X, T) .

Closed Set in a Topology

A subset S of X is said to be closed, if its complement $X - S$ is in T .

Neighbourhood of a Point x , or a Set S , in Topological Space (X, T)

A neighborhood of a point x (or a set S), in a topological space is an open set, i. e, a member of T , which contains the point x (or the set S).

Continuity of a Function in Topological Space

For two topologies (X_i, T_i) , $i = 1, 2$, a function $f: X_1 \rightarrow X_2$ from X_1 to X_2 is said to be *continuous* at a point x_0 of X_1 , if for each neighbourhood H of $f(x_0)$ in X_2 , there exists a neighbourhood G of x_0 in X_1 such that $f(G) \subseteq H$.

Connectedness in topological space

In a topological space (X, T) , *a set $S \in T$ is connected if it cannot be written as disjoint union of two or more non-empty open sets.*

Cover

In a topological space (X, T) , for a set S , **cover of S is a family F of subsets of X** with the property that $S \subset \bigcup_{B \in F} B$.

Open Cover

In a topological space, a cover is called an open cover if every set B in the family F is an open set, *i. e. $F \subseteq T$* .

Subcover and Finite Subcover of a Cover

In a topological space, for a cover F of the set S , a subcover F' is a family—with $F' \subseteq F$ —which is also

Compact Set/ Space

a cover of S . If, in addition, $\mathcal{F}'$ contains only finitely many sets, we call it a **finite** subcover.

A set S in a topological space $(X, \mathcal{T})$ is called compact if every open cover of S has a finite subcover. If X , itself, is compact, we say X is a **compact space**.

15.6 ANSWERS OR HINTS TO CHECK YOUR PROGRESS

Check Your Progress 1

- 1) Show that $(\mathbb{R}^2, d_T)$ is a metric space, where, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$, and d_T , the Taxicab distance, is defined as follows: For $x = (x_1, x_2)$, and $y = (y_1, y_2)$,

$$d_T(x, y) = |x_1 - y_1| + |x_2 - y_2| = \sum_{i=1}^2 |x_i - y_i|, \text{ for } i = 1, 2.$$

- i) As $|x_i - y_i| \geq 0$, for $i = 1, 2$,

$$d_T(x, y) = \sum_{i=1}^2 |x_i - y_i| \geq 0 + 0 = 0$$

- ii) Also, we know, $|x_i - y_i| \geq 0$, for $i = 1, 2$.

$$d_T(x, y) = \sum_{i=1}^2 |x_i - y_i| = 0 \text{ iff } |x_i - y_i| = 0 \text{ for } i = 1, 2.$$

$$\text{iff } x_i - y_i = 0 \text{ (as, } |x| = 0 \text{ iff } x = 0)$$

$$\text{iff } x_i = y_i \text{ for } i = 1, 2$$

$$\text{iff } x = y.$$

- iii) (symmetry)

$$d_T(x, y) = \sum_{i=1}^2 |x_i - y_i| = \sum_{i=1}^2 |y_i - x_i| \text{ (by definition of mod function)}$$

$$= d_T(y, x) \text{ (by definition of } d)$$

- iv) (Triangular Inequality)

For $x = (x_1, x_2)$, $y = (y_1, y_2)$; and $z = (z_1, z_2) \in \mathbb{R}^2$, consider

$$d_T(x, z) = \sum_{i=1}^2 |x_i - z_i| = \sum_{i=1}^2 [| (x_i - y_i) + (y_i - z_i) |]$$

$$\leq \sum_{i=1}^2 [|x_i - y_i| + |y_i - z_i|] \text{ (by } |a_i + b_i| \leq |a_i| + |b_i|).$$

$$= \sum_{i=1}^2 |x_i - y_i| + \sum_{i=1}^2 |y_i - z_i|$$

(by rearranging the terms in the summation)

$$= d_T(x, y) + d_T(y, z)$$

Hence, the proof.

- 2) Show that $(\mathbb{R}^2, d_E)$ is a metric space, where, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$, and d_E , the Euclidean distance, is as defined as follows: For $x = (x_1, x_2)$, and $y = (y_1, y_2)$,

$$d_E(x, y) = \sqrt{[(x_1 - y_1)^2 + (x_2 - y_2)^2]}$$

$$= \sum_{i=1}^2 [(x_i - y_i)^2]$$

Solution:

- i) As $(x_i - y_i)^2 \geq 0$, for $i = 1, 2$,

$$d_E(x, y) = \sum_{i=1}^2 [(x_i - y_i)^2] \geq 0 + 0 = 0$$

- ii) Again as $(x_i - y_i)^2 \geq 0$, for $i = 1, 2$.

Hence,

$$d_E(x, y) = \sum_{i=1}^2 [(x_i - y_i)^2] = 0$$

iff

$$(x_i - y_i)^2 = 0 \text{ for each } i = 1, 2.$$

$$\text{iff } x_i = y_i, \text{ for } i = 1, 2.$$

$$\text{iff } x = y$$

- iii) (symmetry)

$$\begin{aligned} d_E(x, y) &= \sum_{i=1}^2 [(x_i - y_i)^2] = \sum_{i=1}^2 [(y_i - x_i)^2] \\ &= d_T(y, x) \text{ (by definition of } d) \end{aligned}$$

- iv) (Triangular Inequality)

For $x = (x_1, x_2)$, $y = (y_1, y_2)$; and $z = (z_1, z_2) \in \mathbb{R}^2$, consider

$$\begin{aligned} d_E(x, z) &= \sum_{i=1}^2 [(x_i - z_i)^2] \\ &= \sum_{i=1}^2 [(x_i - y_i) + (y_i - z_i)]^2 \end{aligned}$$

(By Pythagorean Theorem, $(a + b)^2 \leq a^2 + b^2$, we get)

$$\begin{aligned} &\leq \sum_{i=1}^2 [(x_i - y_i)^2] + \sum_{i=1}^2 [(y_i - z_i)^2] \\ &= d_E(x, y) + d_E(y, z) \end{aligned}$$

Hence, the proof.

Check Your Progress 2

- 1) For $a \in X$, the open sphere of radius 1,

$$S(a, 1) = \{a\}, \text{ consists of only one point, viz. } a,$$

because, except point } a, \text{ all other points have distance equal to exactly 1,}

but not less than 1. However, } S(a, 1) \text{ consists of all points at a distance less than 1.}

$S^c(a, 1) = X$, because every member of X is at a distance equal to, or less than, 1.

- 2) **The universal set X is an open set:**

Let $x \in X$, then for any $\varepsilon > 0$, the open sphere

$$S(x, \varepsilon) = \{y \in X: d(x, y) < \varepsilon\}, \text{ by definition, } \subseteq X, \text{ the universal set.}$$

Hence, the proof.

- 3) **$\emptyset$, the empty set, is open:** The claim is *proved by contradiction*. Let $\emptyset$ be *not* open. Then $\emptyset$ must contain an element, say y such that $S(y, \varepsilon)$ is not contained in $\emptyset$. But the

assumption about existence of y in $\emptyset$ is contradictory to the fact that $\emptyset$, by definition, has no element.

Hence the proof.

4) **The universal set X is a closed set**

For showing X to be closed, let us consider $X - X$, which equals $\emptyset$, the empty set; and we have proved that $\emptyset$, is an open set. Complement of X is an open set. Hence, X is a closed set.

5) **The empty set, $\emptyset$, is a closed set,**

For showing $\emptyset$ to be closed, let us consider $X - \emptyset = X$, the universal set. We have already proved that X , the universal set, is an open set. Complement of $\emptyset$ is an open set. Hence, $\emptyset$ is a closed set.

Check Your Progress 3

1)

- i) The solutions follow from the theorem which states that a subset of R , equipped with Euclidean metric, is **connected** if and only if **it is an interval**—whether open, semi-open, or closed. Each of N , Z , and Q which is a subset of R , is not an interval in R . And, also, any subset of N , Z , and Q is NOT connected, because each is not an interval in R .
- ii) Any singleton set $\{a\}$ of Euclidean metric space is not connected as $R - \{a\} =] - \infty, a[\cup] a, \infty [$, which is a union of two intervals, each of $] - \infty, a[$ and $] a, \infty [$ is open.
- iii) Any finite subset $\{a_1, a_2, \dots, a_n\}$ of Euclidean metric space is not connected. Let us assume $a_1 < a_2 < \dots < a_n$. Then $R - \{a_1, a_2, \dots, a_n\} =] - \infty, a_1[\cup] a_1, a_2[\cup \dots \cup] a_{n-1}, a_n[\cup] a_n, \infty [$, and each of $] - \infty, a_1[$, $] a_i, a_{i+1}[$, and $] a_n, \infty [$, for $i = 1, 2, \dots, (n-1)$ is an open interval.

2) The first two conditions (i) the universal set $X \in T$, and (ii) the empty set $\emptyset \in T$, by definition of T .

The other two topological conditions are satisfied, as explained below.

It can be easily verified that

- iii) Unions of any number of sets from T is in T , e. g. any union involving X is X , which is in T .

Any union of a set S in T involving $\emptyset$ is S , which is also in T .

Further, $\{a, b\} \cup \{a, c\} = X = \{a, b\} \cup \{a, c\} \cup \{a\} = X$, where X is in T .

- iv) Intersection: any intersection, involving $\emptyset$ is $\emptyset$ which is in T . Any intersection involving

$\{a\}$ is $\{a\}$ is again in T . Further, $\{a, b\} \cap \{a, c\} = \{a\}$ is in T , etc.

- 3) The first condition (i) the universal set $X \in T$ is satisfied as $X - X = \emptyset$, is a finite set.

For the second condition (ii) $\emptyset$ is given, by definition, as a member of T

For the other two conditions, let each of $Y_1, Y_2, \dots$ belong to T .

- i) As union Y of two or more sets say $Y_1, Y_2, \dots$ have more elements than each of the Y_i , hence, Y has more elements than Y_1 . Therefore, $X - Y$ will have lesser number of elements than the number of elements in $X - Y_1$, which is finite. Hence $X - Y$ has only finite number of elements, implying $Y \in T$.
- ii) Also, $X - (\text{Intersection of finitely many } Y_1, Y_2, \dots, Y_n)$, by Demorgan's Laws

$$= \bigcup_{i=1}^n (X - Y_i), \text{ which being finite union of finite sets is finite.}$$

Hence finite intersection of members of T is also in T

- 4) Let $\bigcup_{\alpha} O_{\alpha}$ be an arbitrary, finite or infinite, cover of a finite set $S = \{x_1, \dots, x_k\}$. Then for each x_i , there is some O_{α_i} , with $x_i \in O_{\alpha_i}$, for $1 \leq i \leq k$, and hence, the **finite** collection $\{O_{\alpha_1}, \dots, O_{\alpha_k}\}$ covers E .

15.7 EXERCISES

- 1) **Show that $(\mathbb{R}^n, d_T)$ is a metric space**, where, $\mathbb{R}^n = \mathbb{R} \times \dots \times \mathbb{R}$ $\times$ $\mathbb{R}$, and d_T , the Taxicab distance, is as defined as follows.

For $x = (x_1, x_2, \dots, x_n)$, and $y = (y_1, y_2, \dots, y_n)$

$$\begin{aligned} d_T(x, y) &= d_T(x, y) = |x_1 - y_1| + |x_2 - y_2| + \dots + |x_n - y_n| \\ &= \sum_{i=1}^n |x_i - y_i| \end{aligned}$$

- 2) Show that $(\mathbb{R}^n, d_E)$ is a metric space, where, $\mathbb{R}^n = \mathbb{R} \times \dots \times \mathbb{R} \times \mathbb{R}$, and d_E , the Euclidean distance, is as defined as follows.

For $x = (x_1, x_2, \dots, x_n)$, and $y = (y_1, y_2, \dots, y_n)$

$$d_E(x, y) = \sqrt{\sum_{i=1}^n [(x_i - y_i)^2]}$$

- 3) In the metric space $(\mathbb{R}^2, d_T)$ where, $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$, and d_T , the Taxicab distance, defined as follows, with $d_T(x, y) = |x_1 - y_1| + |x_2 - y_2|$, for $x = (x_1, x_2)$, and $y = (y_1, y_2)$, for $a = (a_1, a_2)$, find $S(a, 1)$, the open sphere and $S^c(a, 1)$, the closed sphere, each of radius 1.

- 4) **Every open sphere $S(a, r)$ is an open set,**

Answers to Exercises

- 1) $d_T(x, y) = \sum_{i=1}^n |x_i - y_i|$

As $|x_i - y_i| \geq 0$, for $i = 1, 2, \dots, n$,

$$d_T(x, y) = \sum_{i=1}^n |x_i - y_i| \geq 0 + 0 + \dots + 0 = 0$$

- i) Again, as $|x_i - y_i| \geq 0$,

$d_T(x, y) = \sum_{i=1}^n |x_i - y_i| = 0$ iff $|x_i - y_i| = 0$ for **each** $i=1, 2, \dots, n$,
iff $x_i - y_i = 0$ for **each** $i=1, 2, \dots, n$ (because, $|x| = 0$, if and only if $x = 0$).
iff $x_i = y_i$ for each $i=1, 2, \dots, n$,
iff $x=y$.

ii) **(symmetry)**

$$d_T(x, y) = \sum_{i=1}^n |x_i - y_i| = \sum_{i=1}^n |y_i - x_i| \text{ (by definition of mod function)} \\ = d_T(y, x) \text{ (by definition of } d)$$

iii) **(Triangular Inequality)**

For $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, $y = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$; and

$z = (z_1, z_2, \dots, z_n) \in \mathbb{R}^n$, consider

$$d_T(x, z) = \sum_{i=1}^n |x_i - z_i| = \sum_{i=1}^n [| (x_i - y_i) + (y_i - z_i) |] \\ \leq \sum_{i=1}^n [|x_i - y_i| + |y_i - z_i|] \text{ (by } |a_i + b_i| \leq |a_i| + |b_i|). \\ = \sum_{i=1}^n |x_i - y_i| + \sum_{i=1}^n |y_i - z_i|$$

(by rearranging the terms in the summation)

$$= d_T(x, y) + d_T(y, z)$$

Hence, the proof.

2)

i) As $(x_i - y_i)^2 \geq 0$, for $i=1, 2, \dots, n$.

$$d_E(x, y) = \sqrt{\sum_{i=1}^n [(x_i - y_i)^2] \geq 0 + 0 + \dots + 0 = 0}$$

ii) Again, as $(x_i - y_i)^2 \geq 0$, for $i=1, 2, \dots, n$.

$$d_E(x, y) = \sqrt{\sum_{i=1}^n [(x_i - y_i)^2] = 0 \text{ iff } (x_i - y_i)^2 = 0 \text{ for } i=1, 2, \dots, n.}$$

$$\text{iff } x_i = y_i, i=1, 2, \dots, n.$$

$$\text{iff } x = y$$

iii) **(symmetry)**

$$d_E(x, y) = \sqrt{\sum_{i=1}^n [(x_i - y_i)^2]} = \sqrt{\sum_{i=1}^n [(y_i - x_i)^2]} \\ = d_E(y, x) \text{ (by definition of } d)$$

iv) **(Triangular Inequality)**

For $x = (x_1, x_2, \dots, x_n)$, $y = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$;

and $z = (z_1, z_2, \dots, z_n) \in \mathbb{R}^n$, consider

$$d_E(x, z) = \sqrt{\sum_{i=1}^n [(x_i - z_i)^2]} \\ = \sqrt{\sum_{i=1}^n [(x_i - y_i) + (y_i - z_i)]^2} \dots (a)$$

(By generalized Pythagorean Theorem,

$$(a_1 + a_2 + \dots + a_n)^2 \leq a_1^2 + a_2^2 + \dots + a_n^2)$$

we get